WEBRUM-05: Error Analysis

> **Series:** WEBRUM — Web Real User Monitoring | **Notebook:** 5 of 10 | **Created:** March 2026 | **Last Updated:** 10/31/2026

Overview

JavaScript errors are among the most impactful issues affecting web application user experience. A single unhandled exception can break page functionality, prevent form submissions, or cause entire features to stop working. Dynatrace RUM captures JavaScript errors, XHR/fetch errors, and custom errors in real-time, enabling you to quantify error impact on real users.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
RUM Enabled	Web applications with error capture active
Permissions	`storage:user.events:read`, `storage:user.sessions:read`
Previous Notebook	WEBRUM-01: RUM Fundamentals

1. Error Types in RUM

Dynatrace captures several categories of browser-side errors:

Error Type	Description	Example
JavaScript error	Unhandled exception or runtime error	`TypeError: Cannot read property 'x' of undefined`
XHR error	Failed XHR/fetch request (HTTP 4xx/5xx or network error)	`HTTP 500 on POST /api/checkout`
Custom error	Errors reported via the RUM API (`dtrum.reportError()`)	`Payment validation failed`
CSP violation	Content Security Policy blocking a resource	`Refused to load script from 'http://evil.com'`
Resource error	Failed resource loading (images, scripts, stylesheets)	`Failed to load /assets/app.js`

Error Data Model

Error events are `user.events` records with `characteristics.has_error == true`:

Field	Description
`error.reason`	The error reason — RUM JavaScript reports a pre-defined set of values

Field	Description
`error.type`	Error classification: `exception`, `request` (failed request), `csp`, `reported` (custom reported error)
`error.source`	The error source, e.g. `fetch`, `console`
`page.detected_name`	Page on which the error occurred
`dt.rum.session.id`	Session containing the error
`start_time`	When the error occurred — `user.events` has no populated `timestamp`

The classic names `error.message`, `action.name` and `sessionId` read null on New RUM data.

> **Sources:** Errors — semantic dictionary (DT docs) — *"The error reason. RUM JavaScript reports a pre-defined set of values."*

```
// Error / navigation vocabulary corrected 08/12/2026 (New RUM):
//   filter type == "Error"  -&gt; filter characteristics.has_error == true
(11,909 events; identical
//                              population to isNotNull(error.type), whose
values are request/csp/exception)
//   error.message           -&gt; error.reason
//   user_action.type: "Load" -&gt; "hard_navigation", "RouteChange" -&gt;
"soft_navigation",
//                     "Xhr" -&gt; "same_view", "Custom" -&gt; "api"
(documented values, corrected 10/05/2026)
//   connection.type         -&gt; no web RUM equivalent
(network.connection.type is OneAgent for Mobile
//                              only; network.protocol.name is the OSI
protocol, e.g. "http")
// THE EVENT CHARACTERISTIC MATTERS AS MUCH AS THE FIELD: navigation-timing
fields
// (performance.dom_interactive, performance.load_event_end) live on
navigation events
// (characteristics.has_navigation) and are unpopulated on page summaries
// (characteristics.has_page_summary), so a page-summary filter silently
empties them.
// ttfb.* is the opposite — it lives on page summaries. Select with the stable
has_* flags, not
// characteristics.classifier (corrected 09/28/2026): the docs call classifier
"not intended for
// query usage" and Semantic Dictionary 1.349 removes it from the user-event
models (re-listed in SD 1.350, still not for query use).
// Session/performance field vocabulary corrected 08/12/2026 (New RUM).
Classic camelCase RUM
// names are null on New RUM data and fail silently. Verified against 3,261
user.sessions:
//   userType -&gt; dt.rum.user_type      userActionCount -&gt;
user_action_count
//   totalErrorCount -&gt; error.count    sessionId -&gt; dt.rum.session.id
//   hasSessionReplay -&gt; characteristics.has_replay
//   browserFamily -&gt; browser.name     osFamily -&gt; os.name
//   application -&gt; frontend.name (stable; an array on user.sessions, so
expand it before grouping)
//                  primary_tags.application exists only if your OpenPipeline
adds it (corrected 10/14/2026)
//   country/city/continent -&gt; geo.country.name / geo.city.name /
geo.continent.name
//   screen.width|height -&gt; browser.window.width|height
//   dom.interactive.time -&gt; performance.dom_interactive
//   load.event.time -&gt; performance.load_event_end
//   server.time -&gt; ttfb.value (page summaries;
web_vitals.time_to_first_byte where populated) —
```

```
//                  NOT ttfb.waiting_duration, which is the pre-request
wait/redirect phase
// TWO TENANT CAVEATS on the validation tenant, both of which leave a CORRECT
query empty:
//   * every session is dt.rum.user_type == "synthetic", so a real-user filter
matches nothing —
//     the documented values are "real_user" / "robot" / "synthetic"
(lowercase);
//   * geo.* is 0-populated, because synthetic traffic carries no geolocation.
// Field vocabulary corrected 08/12/2026 — this series targets **New RUM**,
but was written
// against names that are null on New RUM data, so these cells returned
nothing while erroring
// nowhere. Verified against 5,556,127 user.events records (schema 0.24.0,
javascript agent):
//   action.type == "Load"              -&gt; characteristics.has_navigation
== true
//                                        (was characteristics.classifier —
"not intended for query usage", SD 1.349)
//   action.type                        -&gt; user_action.type
(hard_navigation | soft_navigation | same_view | api)
//   action.name                        -&gt; page.detected_name
//   LCP / CLS                          -&gt;
web_vitals.largest_contentful_paint / web_vitals.cumulative_layout_shift
//                                        (stable — the fields Dynatrace's own
DQL examples use), falling back to
//                                        the experimental lcp.start_time /
cls.value (corrected 10/14/2026)
//   app.name                           -&gt; frontend.name (stable; an array
on user.sessions)
// UNITS DIFFER BETWEEN THE TWO FAMILIES. web_vitals.largest_contentful_paint
is a DURATION, so
// `/ 1ms` turns it into milliseconds; lcp.start_time is a PLAIN NUMBER
already in milliseconds,
// and dividing it by 1ms yields null. The CWV cells therefore read
// coalesce(web_vitals.largest_contentful_paint / 1ms, lcp.start_time) and
// coalesce(web_vitals.cumulative_layout_shift, cls.value), against the
2500/4000 ms and 0.1/0.25 thresholds.
// Why both: on the validation tenant every user event comes from the
synthetic engine, which fills
// lcp.* / cls.* but not web_vitals.* (0 of 243,494 page summaries in 30 days,
10/14/2026). That
// is a property of synthetic-monitor data, not a sign that web_vitals.* is
rarely populated.
// Recent RUM errors — explore the data structure
fetch user.events, from:-1h
| filter characteristics.has_error == true
| fieldsKeep start_time, error.reason, error.type, error.source,
```

```
page.detected_name, frontend.name, dt.rum.session.id
| sort start_time desc   // user.events has no populated timestamp; sorting on
it does not order rows
| limit 20
```

2. Error Volume and Trends

Start by understanding overall error volume and how it changes over time.

```
// Error count by type over last 24 hours
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize error_count = count(), by:{error.type}
| sort error_count desc
```

```
// Error trend over 24 hours — hourly bucketed by error type
fetch user.events, from:-24h
| filter characteristics.has_error == true
| makeTimeseries error_count = count(), interval:1h, by:{error.type}
```

```
// Error volume by application — which apps have the most errors?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize {error_count = count(),
    unique_errors = countDistinct(error.reason),
    affected_sessions = countDistinct(dt.rum.session.id)},
    by:{frontend.name}
| sort error_count desc
```

3. Error Grouping and Top Errors

Error messages often contain variable data (stack traces, URLs, IDs). Grouping by error message helps identify the most frequent issues.

```
// Top 15 errors by frequency — the most common error messages
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize {error_count = count(),
    affected_sessions = countDistinct(dt.rum.session.id)},
    by:{error.reason, error.type}
| sort error_count desc
| limit 15
```

```
// Errors by page — which pages generate the most errors?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter isNotNull(page.detected_name)
| summarize {error_count = count(),
    unique_errors = countDistinct(error.reason)},
    by:{page.detected_name}
| sort error_count desc
| limit 10
```

4. Error Impact Analysis

Not all errors are equal. An error affecting 1 session is different from one affecting 10,000. Impact analysis quantifies the blast radius of each error.

```
// Error impact — errors ranked by number of affected sessions
fetch user.events, from:-24h
| filter characteristics.has_error == true
| summarize {total_occurrences = count(),
    affected_sessions = countDistinct(dt.rum.session.id)},
    by:{error.reason}
| sort affected_sessions desc
| limit 10
```

```
// Error rate per application — percentage of sessions with errors
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| expand frontend.name   // array on sessions: a multi-frontend session counts
once per frontend
| summarize {total_sessions = count(),
    error_sessions = countIf(error.count &gt; 0)},
    by:{frontend.name}
| fieldsAdd error_rate_pct = round(toDouble(error_sessions) /
toDouble(total_sessions) * 100.0, decimals: 2)
| sort error_rate_pct desc
```

> **Tip:** Focus on errors with high session impact rather than high occurrence count. A single user hitting a retry loop can generate thousands of occurrences from one session.

5. XHR and Fetch Errors

XHR/fetch errors indicate backend API failures impacting the frontend. These are particularly critical in SPAs where the UI depends entirely on API responses.

```
// Unit trap (08/12/2026): New RUM timing fields such as lcp.start_time and
ttfb.value
// are PLAIN NUMBERS already in milliseconds, not durations. `field / 1ms`
yields null on them —
// silently, so a chart of nulls looks like "no data". Compare and aggregate
them directly.
// XHR errors by action — identify failing API calls
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter error.type == "request"
| summarize {error_count = count(),
    affected_sessions = countDistinct(dt.rum.session.id)},
    by:{error.reason, page.detected_name}
| sort error_count desc
| limit 15
```

```
// XHR error trend — are backend errors increasing?
fetch user.events, from:-24h
| filter characteristics.has_error == true
| filter error.type == "request"
| makeTimeseries error_count = count(), interval:1h
```

6. Rage Click Detection

A **rage click** occurs when a user rapidly clicks the same element multiple times — a strong signal of frustration. This usually indicates:

- A button that appears clickable but is not responding
- A form submission that is silently failing
- A UI element that is loading too slowly
- A dead link or broken navigation element

RUM Classic records rage clicks on the user session (USQL `numberOfRageClicks`). **New RUM documents no rage-click field.** The Semantic Dictionary has no field named for rage clicks or frustration, and `type` — which older versions of this notebook filtered as `type == "RageClick"` — is the *entity* type, empty on user events, so that filter returned nothing on every tenant. Until Dynatrace documents a New RUM rage-click signal, use Session Replay (WEBRUM-07) and the error-session correlation below to find frustrated users.

> **Dictionary:** no field matching `contains(name, "rage_click")`, `"rageclick"`, `"rage.click"` or `"frustrat"` in `dt.semantic_dictionary.fields`, read 10/05/2026 (control: the same query returned `network.connection.type` and `network.connection.subtype`); `type` (`stable`) — *"The entity type."*

7. Error-to-Session Correlation

Understanding how errors correlate with session outcomes (bounce, low engagement, failed conversion) helps prioritize fixes.

```
// Compare sessions with and without errors — engagement impact
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| fieldsAdd has_errors = if(error.count &gt; 0, "With Errors", else: "No
Errors")
| summarize {session_count = count(),
    avg_actions = avg(user_action_count),
    avg_duration_sec = avg(duration / 1s),
    bounce_rate = countIf(user_action_count == 1)},
    by:{has_errors}
| fieldsAdd bounce_pct = round(toDouble(bounce_rate) / toDouble(session_count)
* 100.0, decimals: 1)
```

```
// Errors by browser — are certain browsers more error-prone?
fetch user.sessions, from:-24h
| filter dt.rum.user_type == "real_user"
| filter isNotNull(browser.name)
| summarize {total = count(),
    with_errors = countIf(error.count &gt; 0)},
    by:{browser.name}
| fieldsAdd error_rate = round(toDouble(with_errors) / toDouble(total) *
100.0, decimals: 1)
| filter total &gt; 10
| sort error_rate desc
| limit 10
```

8. Summary and Next Steps

In this notebook, we covered:

- **Error types** — JavaScript errors, XHR errors, custom errors, resource errors
- **Error volume and trends** — Tracking error rates over time
- **Error grouping** — Identifying the most frequent error messages
- **Impact analysis** — Measuring how many sessions each error affects
- **XHR/fetch errors** — Backend API failures visible from the frontend
- **Rage clicks** — a RUM Classic signal with no documented New RUM field; use Session Replay instead
- **Error-session correlation** — How errors impact engagement and bounce rates

Next Steps

- **WEBRUM-06: Performance Analysis** — Page load waterfall and performance bottleneck identification
- **WEBRUM-07: Session Replay** — Visually investigate error-impacted sessions

References

- Configure error detection for web applications (DT docs)
- User actions in RUM Classic (DT docs)
- User events — semantic dictionary (DT docs) — *"Used for internal optimization when storing the data and not intended for query usage."*
- Semantic Dictionary changelog 1.349 (DT docs)

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